

WASP-39b

R-2148

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-39_120313 · created 2026-08-12T00:53:32+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2455999.927 +/- 0.021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.135 +/- 0.023
Transit depth (Rp/Rs)^2	1.83 +/- 0.63 [%]
Orbital Inclination (inc)	87.89 +/- 0.95
Airmass coefficient 1 (a1)	0.973 +/- 0.011
Airmass coefficient 2 (a2)	0.02 +/- 0.042
Scatter in the residuals of the lightcurve fit is	1.16 %
Best Comparison Star	#1 - [220, 394]
Optimal Method	PSF photometry
Transit Duration (day)	0.112 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.135 ± 0.023 vs 0.1457 (deviation 0.46σ)
Duration fit vs published	0.112 d vs 0.12 d (deviation 0.66σ)
Mid-time O–C	2.9 min
Depth SNR	5.8
Residual scatter	1.16 %

Light curve

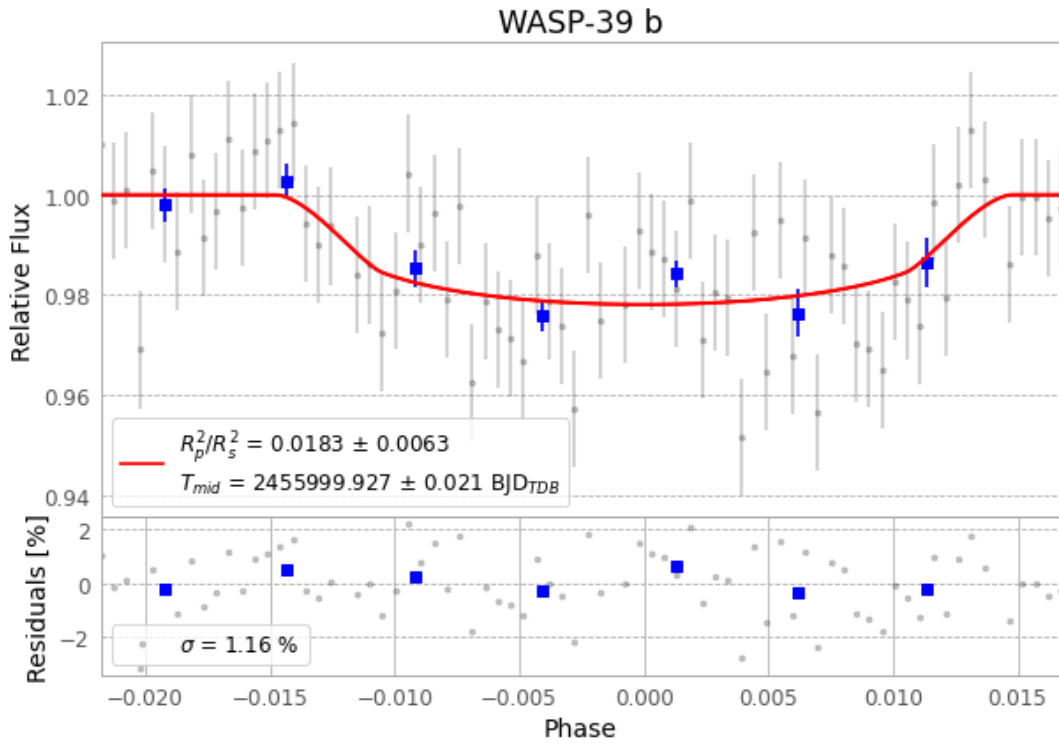


Figure 1 — FinalLightCurve_WASP-39 b_2012-03-13.png (SHA-256 00b0f747feaf6b4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [309, 264], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4f461c91bc934e92...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

76 FITS frames (50 MB), WASP-39120313080012.FITS → WASP-39120313114512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-39-120313.log (ab83789472d3...), FinalLightCurve_WASP-39 b_2012-03-13.png (00b0f747feaf6b4...), FinalLightCurve_WASP-39 b_2012-03-13.csv (1929f3bebd0...)

Compute resources

Wall time 172.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-12 00:53Z.